UNIVERSIDAD CATÓLICA BOLIVIANA
LA PAZ

Carrera de Ingeniería Mecatrónica

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EMBEDDED SYSTEMS II

AGENDA

1. Bit Hex Unpack
2. TIVA GPIO configuration
3. TIVA Timers
4. Tiva ADC

BIT HEX UNPACK

- Is the process of manage addresses of different GPIO.
- In this case we will use this for TIVA GPIO definition.
- We can use the following process:

BIT HEX UNPACK

0x00 → First pack of bits

↓ Second pack of bits

0x00 = 0000 0000

BIT HEX UNPACK

0x00 → First pack of bits

↓ Second pack of bits

Bit 0 to 7

0x03 = 0000 0011

Enables pin 0 and 1

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IMT UCB

BIT HEX UNPACK

- As we saw each bit represents a pin value going from 0 to 7.
- We have to read the values from RIGHT to LEFT
- The max value would be 0xff.

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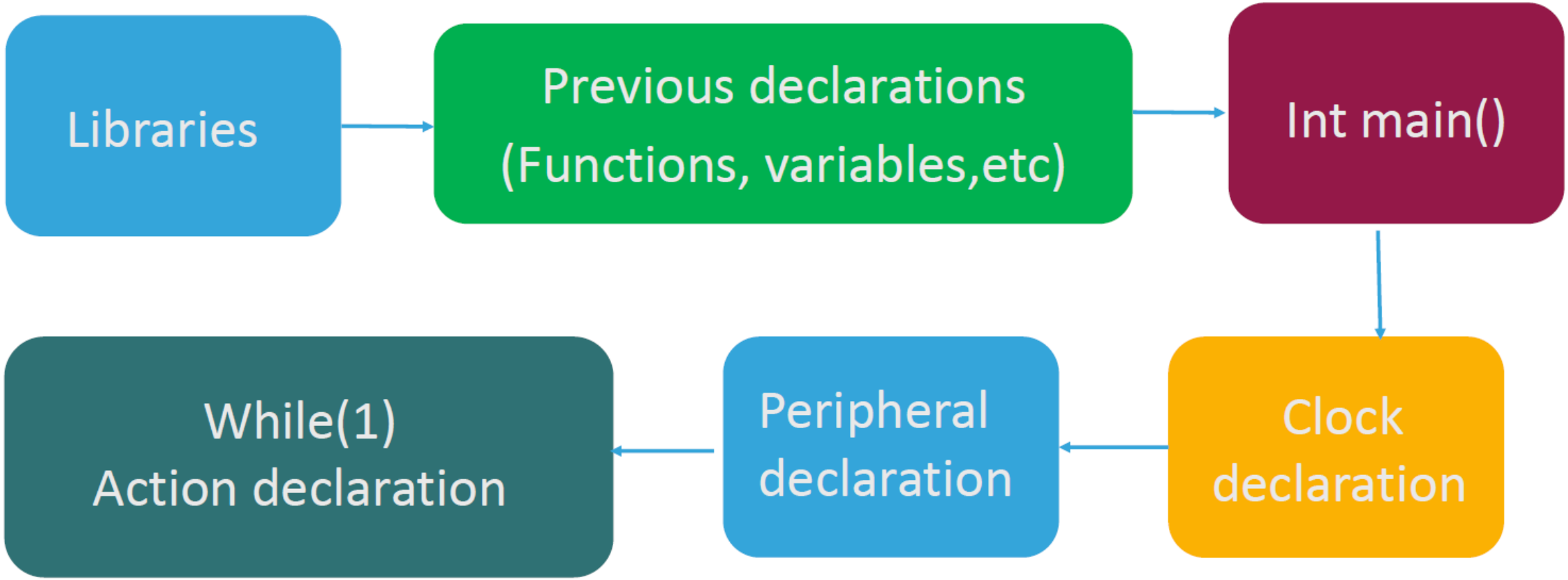
Questions?

PROJECT IN TIVA



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TIVA GPIO CONFIGURATION

- TIVA is especially complicate to understand and learn in comparison to some previous boards.
- However, with the right tools we can improve our skills
- We need: Driverlib, TIVA datasheet, TIVA peripheral driver library, TIVA examples.

TIVA GPIO CONFIGURATION

- TIVA can use General Purpose Input/Output for different uses like:
  - Read sensor values
  - Send output signal to some pins
  - For read values we use an ADC peripheral
  - For write values we can send digital signals or use PWM
  - Let’s see how to manage a GPIO digital output

TIVA GPIO CONFIGURATION

```
//Libraries needed for TIVA functions
#include <stdint.h>
#include <stdbool.h>
#include "inc/hw_memmap.h"
#include "driverlib/debug.h"
#include "driverlib/gpio.h"
#include "driverlib/sysctl.h"
```

Libraries

```
//*****************************************************************************
//
// The error routine that is called if the driver library encounters an error.
//
//*****************************************************************************
#ifdef DEBUG
void
__error__(char *pcFilename, uint32_t ui32Line)
{
    while(1);
}
#endif

int main(void)
{
    volatile uint32_t ui32Loop;

    SysCtlClockFreqSet((SYSCTL_XTAL_25MHZ | SYSCTL_OSC_MAIN | SYSCTL_USE_PLL | SYSCTL_CFG_VCO_480),120000000); // Eables system clock

    SysCtlPeripheralEnable(SYSCTL_PERIPH_GPION); //Enables peripheral N
    SysCtlPeripheralEnable(SYSCTL_PERIPH_GPIOF); //Enables peripheral F
```

Clock configuration

Peripheral enabling

PROJECT IN TIVA

```
SysCtlPeripheralEnable(SYSCTL_PERIPH_GPION); //Enables peripheral N
SysCtlPeripheralEnable(SYSCTL_PERIPH_GPIOF); //Enables peripheral F

// Check if the peripheral access is enabled.
while(!SysCtlPeripheralReady(SYSCTL_PERIPH_GPION))
{
}

while(!SysCtlPeripheralReady(SYSCTL_PERIPH_GPIOF))
{
}

GPIOPinTypeGPIOOutput(GPIO_PORTN_BASE, 0x03); //enables pin 0 and 1
GPIOPinTypeGPIOOutput(GPIO_PORTF_BASE, 0x11); //enables pin 0 and 4
```

Pin enabling (using hex direction)

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PROJECT IN TIVA

Program execution

```
    // Loop forever.
    while(1)
    {
        // Turn on the LEDs.
        GPIOPinWrite(GPIO_PORTN_BASE,0x03,0x02);  //set periph base,pins, values
        GPIOPinWrite(GPIO_PORTF_BASE,0x11,0x01);  //set periph base,pins, values

        // Delay for a bit.
        for(ui32Loop = 0; ui32Loop < 20000000; ui32Loop++)
        {
        }

        // Turn off the LEDs.
        GPIOPinWrite(GPIO_PORTN_BASE,0x03,0x01);
        GPIOPinWrite(GPIO_PORTF_BASE,0x11,0x10);

        // Delay for a bit.
        for(ui32Loop = 0; ui32Loop < 20000000; ui32Loop++)
        {
        }

    }
}
```

PROJECT IN TIVA

Program execution

```
//Set GPIO as input/output
GPIOPinTypeGPIOOutput(GPIO_PORTN_BASE, 0x03); //enables pin 0 and 1
GPIOPinTypeGPIOOutput(GPIO_PORTF_BASE, 0x11); //enables pin 0 and 4
//GPIO as Input
GPIOPinTypeGPIOInput(GPIO_PORTJ_BASE, 0x00);
//Configures input current strenght and pull type
GPIOPadConfigSet(GPIO_PORTJ_BASE,GPIO_PIN_0|GPIO_PIN_1,GPIO_STRENGTH_2MA, GPIO_PIN_TYPE_STD_WPU);
```

As you can see, we use the word Input, and we need to declare the pull type (see documentation)

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PROJECT IN TIVA

Program execution

```
if (GPIOPinRead(GPIO_PORTJ_BASE, GPIO_PIN_0) == 0)
 {
  button ++;
 }
```

We can read the value of the input depending of the pull type (pull up = 0 and pull down= GPIO)

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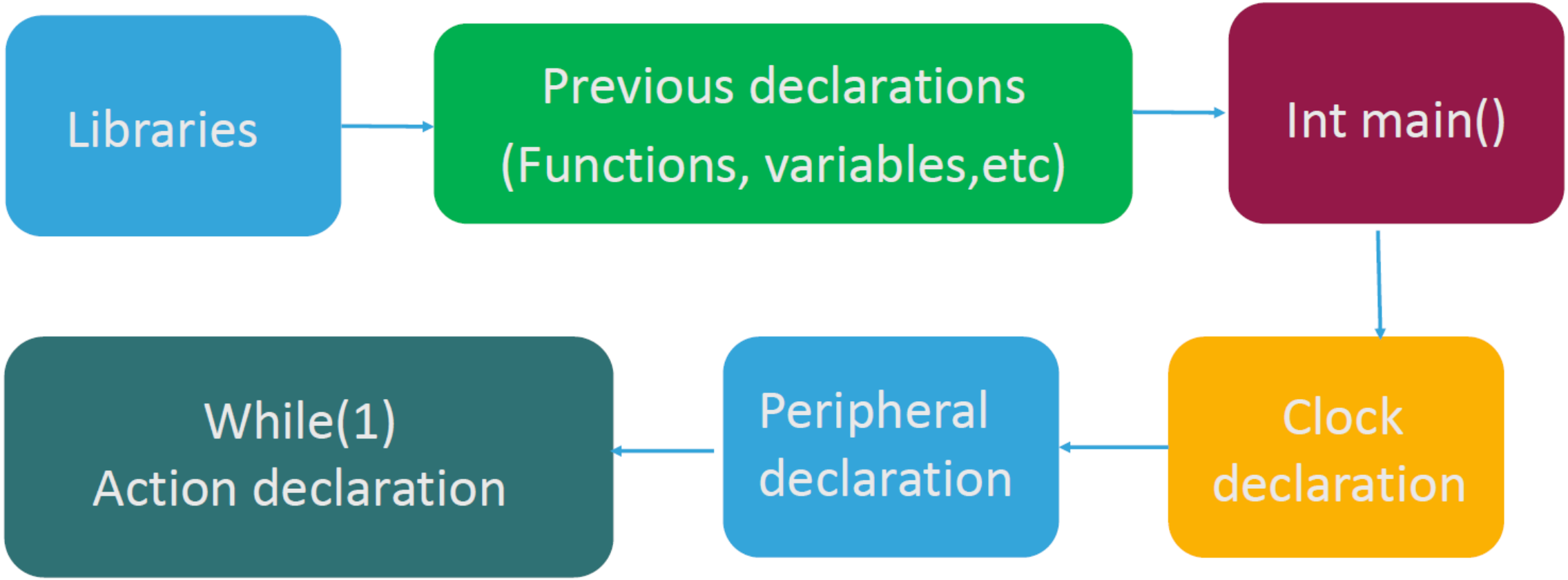
PROJECT IN TIVA



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TIVA GPIO CONFIGURATION

- Is necessary to enable the peripheral for the GPIO.
- We define the pin type as input or output (in this case output)
- As we are using outputs we have to write the pin value.
- The value to write is the GPIO value (NOT 1) or 0.

TIVA TIMERS

- Timers are used to generate operations according to some events during the time.
- Generally, we will generate interruptions to the main routine when a timer is executed.
- We can use delay to interrupt all the program execution and timers to change the main routine for a sub routine.

TIVA TIMERS

```
uint32_t FS= 120000000*2;
void timer0A_handler(void);
uint8_t switch_state = 0;
uint32_t button = 0;
int main(void)
{
    //Clock Configuration
    SysCtlClockFreqSet((SYSCTL_XTAL_25MHZ | SYSCTL_OSC_MAIN | SYSCTL_USE_PLL | SYSCTL_CFG_VCO_480),120000000); // Eables system clock
    //Enable peripherals
    SysCtlPeripheralEnable(SYSCTL_PERIPH_GPION); //Enables peripheral N
    SysCtlPeripheralEnable(SYSCTL_PERIPH_GPIOF); //Enables peripheral F
    SysCtlPeripheralEnable(SYSCTL_PERIPH_GPIOJ);
    //Enable Timer Peripheral
    SysCtlPeripheralEnable(SYSCTL_PERIPH_TIMER0);
```

We must enable the Timer peripheral to use it.

PROJECT IN TIVA

```
//Set timer
TimerConfigure(TIMER0_BASE,TIMER_CFG_PERIODIC);
//Set the cout time for Timer
TimerLoadSet(TIMER0_BASE, TIMER_A, FS);
//Enable processor interrupts
IntMasterEnable();
//Enable Interrupt
IntEnable(INT_TIMER0A);
//Enable timer A interrupt
TimerIntEnable(TIMER0_BASE, TIMER_TIMA_TIMEOUT);
//Enable the timer
TimerEnable(TIMER0_BASE,TIMER_A);
```

This is the main routine to deine a timer, configure it and enable it.

FS is the frequence sample, it is used to get the timer intervals:

timerFreq=clockfreq/FS

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PROJECT IN TIVA

```
void timer0A_handler(void)
{
    switch_state ++;
    //Changing the timer delay
    if (GPIOPinRead(GPIO_PORTJ_BASE, GPIO_PIN_0) == 0)
     {
      button ++;
     }
    if(button >0)
    {
          FS =120000000*4;
          TimerLoadSet(TIMER0_BASE, TIMER_A, FS);
    }
    //Clear timer
    TimerIntClear(TIMER0_BASE, TIMER_A);
    // Sequence 1.
    if(switch_state % 2 != 0)
    {
         GPIOPinWrite(GPIO_PORTN_BASE,0x03,0x02);  //set periph base,pins, values
         GPIOPinWrite(GPIO_PORTF_BASE,0x11,0x01);  //set periph base,pins, values
    }
    // Sequence 2.
    if(switch_state % 2 == 0)
    {
          GPIOPinWrite(GPIO_PORTN_BASE,0x03,0x01);
```

The timer uses a external function to execute the sub routine. This routine will stop the main routine and depending if the timer is periodic or one shot will keep executing

PROJECT IN TIVA

```
//*****************************************************************************
//
// The entry point for the application.
//
//*****************************************************************************
extern int main(void);
extern void timer0A_handler(void);
//*****************************************************************************
```

We have to declare the timer function in the startup file of the project.

```
3    IntDefaultHandler,                    // ADC Sequence 0
4    IntDefaultHandler,                    // ADC Sequence 1
5    IntDefaultHandler,                    // ADC Sequence 2
6    IntDefaultHandler,                    // ADC Sequence 3
7    IntDefaultHandler,                    // Watchdog timer
8    timer0A_handler,                    // Timer 0 subtimer A
9    IntDefaultHandler,                    // Timer 0 subtimer B
0    IntDefaultHandler,                    // Timer 1 subtimer A
1    IntDefaultHandler,                    // Timer 1 subtimer B
```

THANK YOU FOR YOUR ATTENTION!